Leave One Out: Recap

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Module 3: Resample, Session 2

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The jackknife SE

$T_{(i)}$ is the statistic computed without x_i, using `x[-i]`.

$$\widehat{\mathrm{SE}}_{\mathrm{jack}}(T) = \sqrt{\frac{n-1}{n}\sum_{i=1}^{n}\left(T_{(i)} - \bar{T}_{(\cdot)}\right)^2}, \qquad \bar{T}_{(\cdot)} = \frac{1}{n}\sum_{i=1}^{n} T_{(i)}.$$

- For the mean, the sample variance of the $T_{(i)}$ is only $s^2/(n-1)^2$. The factor $(n-1)/n$ makes the SE exactly $s/\sqrt{n}$.
- n evaluations and no Monte Carlo error. The bootstrap takes B evaluations.
- Flint, correlation of logged first draw and flushed: jackknife SE 0.0437, bootstrap SE 0.0432.

Failure case: the median

- The leave-one-out medians take two values, $x_{(m)}$ and $x_{(m+1)}$ for $n = 2m$, or three for odd n.
- So the jackknife SE depends only on the gap between the two middle observations.
- MDEQ: 11 of 71 samples equal the median, so every $T_{(i)}$ is 3 ppb and the jackknife SE is 0.
- For normal data the ratio of jackknife to true SE tends to an Exponential(1), not to 1.
- The jackknife needs a smooth T: a mean, variance, correlation or regression coefficient. For quantiles, bootstrap.

The bootstrap-t interval

Studentize each bootstrap replicate with an SE from **its own** bootstrap sample:

$$W_b^* = \frac{T_b^* - T}{\widehat{\mathrm{SE}}_b^*}, \qquad \left[T - \widehat{\mathrm{SE}}\, W_{1-\alpha/2}^*,\ T - \widehat{\mathrm{SE}}\, W_{\alpha/2}^* \right].$$

- A nested bootstrap costs $B \times B_2$ evaluations. A jackknife in each sample costs $B \times n$.

Flint correlation, $r = 0.648$	95% interval
percentile	[0.557, 0.727]
bootstrap-t	[0.547, 0.725]

- The SE of a correlation shrinks as it grows, so the bootstrap-t reaches further down.

Cross-validation

- **Training error** scores $\hat{f}$ on the points that chose it, so it is optimistic.
- **Leave-one-out**: average $L(Y_i, \hat{f}_{(i)}(X_i))$. For least squares, one fit gives every $e_i/(1 - h_{ii})$.
- K-**fold**: fit to $K - 1$ random folds, predict the one left out. Use the same folds for every model.

Flint, `poly(lx, d)`	$d = 1$	$d = 5$	$d = 10$
training MSE	1.266	1.240	1.216
leave-one-out MSE	1.288	1.323	1.688

Choosing a loss

With $e = y - \hat{y}$, a constant prediction c that minimizes each loss is

loss	$L(e)$	best c
squared	e^2	mean
absolute	$\lvert e \rvert$	median
pinball	τe if $e > 0$, $(1-\tau)(-e)$ if $e \leq 0$	τ quantile

- With $\tau = 0.9$, predicting one unit too low costs 9 times as much as one unit too high.
- Write down what each kind of error costs, and use that loss. Cross-validation works with any loss.